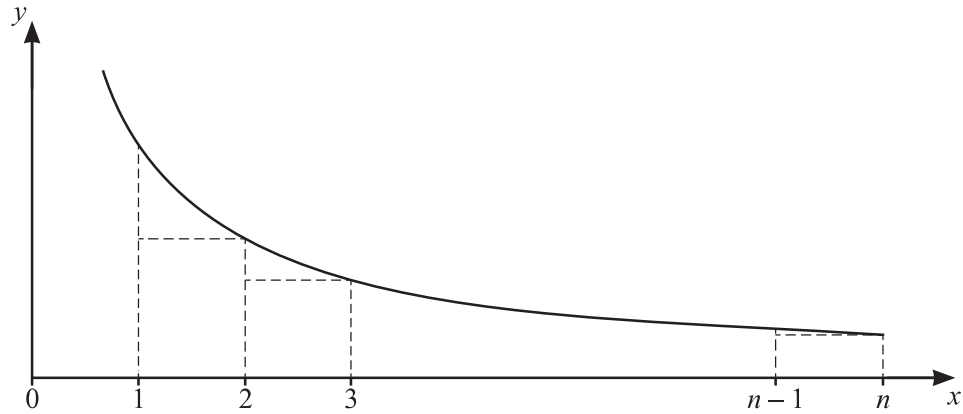




The diagram shows the curve $y = \frac{1}{\sqrt{x^2 + 2x}}$ for $x > 0$, together with a set of $(n-1)$ rectangles of unit width.

By considering the sum of the areas of these rectangles, show that

$$\sum_{r=1}^n \frac{1}{\sqrt{r^2+2r}} < \ln(n+1+\sqrt{n^2+2n}) + \frac{1}{3}\sqrt{3} - \ln(2+\sqrt{3}). \quad [10]$$

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.